Slide 1: Introduction to Python Arrays

- An array stores multiple values in one variable.
- All elements usually have the same data type.
- In Python, we commonly use lists as arrays.
- Example: numbers = [10, 20, 30, 40]

Slide 2: Creating an Array

- Using List: arr = [1, 2, 3, 4, 5]
- Using array module:
- import array
- arr = array.array('i', [1, 2, 3, 4])
- 'i' means integer type.

Slide 3: Accessing Elements

- Index starts from 0.
- Example:
- arr = [10, 20, 30, 40]
- print(arr[0]) # 10
- print(arr[2]) # 30

Slide 4: Updating Elements

- arr = [10, 20, 30]
- arr[1] = 50
- Output: [10, 50, 30]

Slide 5: Adding Elements

index

val

- append() → Add at end: $\boxed{\text{arr.append(60)}}$
- insert() → Add at specific position: arr.insert(1, 25)

0 1 2

[0, 2, 4]

[0, 2, 4, 60]

[0, 25, 2, 4]

[0, 25, 2, 4]

Slide 6: Removing Elements

- remove() → Remove by value: arr.remove(20)
- pop() → Remove by index: arr.pop(1)

[0, 20, 1, 5]

[0, , 1, 5]

[0, 1, 5]

Slide 7: Array Length

- Use len(arr) to get total number of elements.

Slide 8: Looping Through Array

- for item in arr:
- print(item)

for i, item in arr:

Slide 9: Basic Array Operations

- sum(arr) → Sum of elements
- max(arr) → Maximum value
- min(arr) → Minimum value

Slide 10: Sorting Array

- arr.sort() → Sort in ascending order
- arr.sort(reverse=True) → Sort in descending order

[1, 0, 5, 2]

[0 1 2 5]

5 2 1 0